

# YINMIN ZHONG

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## EDUCATION

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Peking University  
Ph.D. in Computer Science.  
Advisor: Xin Jin

*Sep 2022 - Present*

Peking University  
B.S. in Computer Science.

*Sep 2018 - June 2022*  
GPA: 3.79/4.0 (top 5%)

## RESEARCH INTEREST

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My research lies at the intersection of deep learning and distributed systems, focusing on building production-grade infrastructures for large-scale LLM training and serving. I leverage system-level insights to optimize throughput, latency, and resource efficiency across the entire stack.

## PUBLICATIONS

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**RLHFuse: Efficient RLHF Training for Large Language Models with Inter- and Intra-Stage Fusion**

Yinmin Zhong, Zili Zhang, Bingyang Wu, Shengyu Liu, Yukun Chen, Changyi Wan, Hanpeng Hu, Lei Xia, Ranchen Ming, Yibo Zhu, Xin Jin

USENIX Symposium on Networked Systems Design and Implementation (NSDI 2025)

**DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving**

Yinmin Zhong, Shengyu Liu, Junda Chen, Jianbo Hu, Yibo Zhu, Xuanzhe Liu, Xin Jin, Hao Zhang

USENIX Symposium on Operating Systems Design and Implementation (OSDI 2024)

**Fast Distributed Inference Serving for Large Language Models**

Bingyang Wu\*, Yinmin Zhong\*, Zili Zhang\*, Gang Huang, Xuanzhe Liu, Xin Jin

(\*Equal Contribution)

USENIX Symposium on Networked Systems Design and Implementation (NSDI 2026)

**Scaling Large Language Model Training to More Than 10,000 GPUs**

Ziheng Jiang\*, Haibin Lin\*, Yinmin Zhong\*, Qi Huang, Yangrui Chen, Zhi Zhang, Yanghua Peng, Xiang Li, Cong Xie, Shibiao Nong, Yulu Jia, Sun He, Hongmin Chen, Zhihao Bai, Qi Hou, Shipeng Yan, Ding Zhou, Yiyao Sheng, Zhuo Jiang, Haohan Xu, Haoran Wei, Zhang Zhang, Pengfei Nie, Leqi Zou, Sida Zhao, Liang Xiang, Zherui Liu, Zhe Li, Xiaoying Jia, Jianxi Ye, Xin Jin, Xin Liu

(\*Equal Contribution)

USENIX Symposium on Networked Systems Design and Implementation (NSDI 2024)

**AlpaServe: Statistical Multiplexing with Model Parallelism for Deep Learning Serving**

Zhuohan Li\*, Lianmin Zheng\*, Yinmin Zhong\*, Vincent Liu, Ying Sheng, Xin Jin, Yanping Huang, Zhifeng Chen, Hao Zhang, Joseph E. Gonzalez, Ion Stoica (\*Equal Contribution)

USENIX Symposium on Operating Systems Design and Implementation (OSDI 2023)

**Towards Efficient Reward Service for RLVR with Request-Level Flexibility and Batch-Level Constraint**

Ruidong Zhu, Mingcong Han, Yinmin Zhong, Gang Huang, Xuanzhe Liu, Xin Jin

USENIX Symposium on Networked Systems Design and Implementation (NSDI 2026)

### **Efficient RL for LLMs with Dynamic and Online Speculative Decoding**

Chao Jin, **Yinmin Zhong**, Zili Zhang, Yimin Jiang, Yibo Zhu

**The 1st Frontier AI Systems Workshop (FAISys 2025)**

### **DistTrain: Addressing Model and Data Heterogeneity with Disaggregated Training for Multimodal Large Language Models**

Zili Zhang, **Yinmin Zhong**, Ranchen Ming, Hanpeng Hu, Jianjian Sun, Zheng Ge, Yibo Zhu, Xin Jin

**ACM International Conference on Applications, Technologies, Architectures, and Protocols for Computer Communication (SIGCOMM 2025)**

### **LoongServe: Efficiently Serving Long-Context Large Language Models with Elastic Sequence Parallelism**

Bingyang Wu, Shengyu Liu, **Yinmin Zhong**, Peng Sun, Xuanzhe Liu, Xin Jin

**ACM Symposium on Operating Systems Principles (SOSP 2024)**

### **Aquifer: Transparent Microsecond-scale Scheduling for vRAN Workloads**

Yunshan Jia, **Yinmin Zhong**, Meng Wang, Jiaqi Gao, Pengyu Zhang, Xuanzhe Liu, Xin Jin

**IEEE Transactions on Services Computing (TSC 2024)**

### **DistMind: Efficient Resource Disaggregation for Deep Learning Workloads**

Xin Jin, Zhihao Bai, Zhen Zhang, Yibo Zhu, **Yinmin Zhong**, Xuanzhe Liu

**IEEE/ACM Transactions on Networking (TON 2024)**

### **ElasticFlow: An Elastic Serverless Training Platform for Distributed Deep Learning**

Diandian Gu, Yihao Zhao, **Yinmin Zhong**, Yifan Xiong, Zhenhua Han, Peng Cheng, Fan Yang, Gang Huang, Xin Jin, Xuanzhe Liu

**ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2023)**

### **StreamRL: Scalable, Heterogeneous, and Elastic RL for LLMs with Disaggregated Stream Generation**

**Yinmin Zhong**, Zili Zhang, Xiaoniu Song, Hanpeng Hu, Chao Jin, Bingyang Wu, Nuo Chen, Yukun Chen, Yu Zhou, Changyi Wan, Hongyu Zhou, Yimin Jiang, Yibo Zhu, Daxin Jiang

**In preprint**

### **UltraEP: Unleash MoE Training and Inference on Rack-Scale Nodes with Near-Optimal Load Balancing**

Xinming Wei, Chao Jin, Tuo Dai, **Yinmin Zhong**, Shan Yu, Chengxu Yang, Bingyang Wu, Zili Zhang, Jing Mai, Qianchao Zhu, Zhouyang Li, Yuliang Liu, Guojie Luo

**In preprint**

### **ReLibra: Routing-Replay-Guided Load Balancing for MoE Training in Reinforcement Learning**

Chao Jin, Xinming Wei, **Yinmin Zhong**, Chengxu Yang, Bingyang Wu, Ruidong Zhu, Zili Zhang, Yuliang Liu, Xin Jin

**In preprint**

### **Heddle: A Distributed Orchestration System for Agentic RL Rollout**

Zili Zhang, **Yinmin Zhong**, Chengxu Yang, Chao Jin, Bingyang Wu, Xinming Wei, Yuliang Liu, Xin Jin

**In preprint**

### **TokenLake: A Unified Segment-level Prefix Cache Pool for Fine-grained Elastic Long-Context LLM Serving**

Bingyang Wu, Zili Zhang, **Yinmin Zhong**, Guanzhe Huang, Yibo Zhu, Xuanzhe Liu, Xin Jin

**In preprint**

## EXPERIENCE

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|-----------------------------------------|-----------------------|
| DeepSeek RL Infra Team                  | April 2025 - Present  |
| · Core System R&D Engineer              |                       |
| StepFun System Team                     | May 2024 - April 2025 |
| Research Intern                         |                       |
| · Mentor: Ranchen Ming, Yibo Zhu        |                       |
| ByteDance AML Team                      | Aug 2023 - May 2024   |
| Research Intern                         |                       |
| · Mentor: Ziheng Jiang, Xin Liu         |                       |
| Peking University                       | Sep 2022 - Present    |
| Research Assistant                      |                       |
| · Advisor: Xin Jin                      |                       |
| Alibaba DAMO Academy                    | Sep 2021 - Sep 2022   |
| Research Intern                         |                       |
| · Mentor: Pengyu Zhang                  |                       |
| AI Innovation Center, Peking University | Sep 2020 - Mar 2021   |
| Software Engineer Intern                |                       |
| · Mentor: Ming Lei                      |                       |

## PROJECTS

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### *Large Language Model Serving*

#### **LoongServe**

**May 2024**

LoongServe proposes a new parallelism paradigm, *elastic sequence parallelism* (ESP), to elastically adapt to the variance between different requests and phases of LLM generation. It (1) improves computation efficiency by elastically adjusting the degree of parallelism in real-time, (2) improves communication efficiency by reducing key-value cache migration overhead and overlapping partial decoding communication with computation, and (3) improves GPU memory efficiency by reducing key-value cache fragmentation across instances.

#### **DistServe**

**Jan 2024**

DistServe improves the performance of large language models (LLMs) serving by disaggregating the prefill and decoding computation. Given the application latency requirements, DistServe co-optimizes the resource allocation and parallelism strategy tailored for each phase.

#### **FastServe**

**April 2023**

Existing LLM serving systems use run-to-completion processing for inference jobs, which suffers from head-of-line blocking and long JCT. FastServe exploits the autoregressive pattern of LLM inference to enable preemption at the granularity of each output token and uses preemptive scheduling to minimize JCT with a novel skip-join Multi-Level Feedback Queue scheduler.

#### **AlpaServe**

**Oct 2022**

AlpaServe targets the multi-tenancy setting for LLM serving and determines an efficient strategy for placing and parallelizing collections of large deep learning models across a distributed cluster. It will trade-off between the overhead introduced by model parallelism and the opportunity to exploit statistical multiplexing to reduce serving latency in the presence of bursty workloads.

## *Deep Learning Training*

### **RLHFuse (deployed in StepFun)**

**Sep 2024**

RLHFuse breaks the traditional view of RLHF workflow as a composition of individual tasks, splitting each task into more finer-grained subtasks. From this new point of view, it opens up the opportunities to fuse the task execution in the RLHF training workflow and greatly improve the training throughput.

### **MegaScale (deployed in ByteDance)**

**Sep 2023**

Optimus is a production system for training large language models (LLMs) at the scale of more than 10,000 GPUs. Optimus takes a full-stack approach that co-designs the algorithmic and system components across operator optimization, model block, optimizer design, computation, communication overlapping, data pipeline, and network performance tuning. We develop a set of diagnosis tools to monitor system components and events deep in the stack, identify root causes, and derive effective techniques to achieve fault tolerance and mitigate stragglers.

### **Alpa**

**Mar 2022**

Alpa automates model-parallel training of large deep learning (DL) models by generating execution plans that unify data, operator, and pipeline parallelism. Alpa makes it simple to train and serve large models like GPT-3 by adding a few lines of code.

### **ElasticFlow**

**Sep 2021**

ElasticFlow is an elastic serverless training platform for distributed deep learning. ElasticFlow provides a serverless interface with two distinct features: (i) users specify only the deep neural network (DNN) model and hyperparameters for a job, but not the number of GPUs; (ii) users specify the deadline for a job, but not the amount of time to occupy GPUs. In contrast to existing server-centric platforms, ElasticFlow provides performance guarantees in terms of meeting deadlines while alleviating tedious, low-level, and manual resource management for deep learning developers.

## *Side Projects*

### **TacOS (developed as course project at Peking University)**

**Feb 2024**

TacOS is an educational Operating System implemented in Rust. It borrows many design philosophies from PintOS, which is a C project adopted by many Operating System courses in the top universities. We provide detailed documentation, comprehensive test cases, and debugging tools for students to accomplish a series of labs based on the TacOS skeleton code.

### **csdiy.wiki**

**Nov 2021**

csdiy.wiki collects various learning materials and online courses in computer science (CS) from top universities in the world. It covers almost all the areas in CS and provides comprehensive guidance for anyone who would like to self-learn CS. This project has earned over 71k stars on GitHub and has repeatedly ranked as the top popular repository in GitHub Trending.

## **TEACHING**

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**Head TA, Operating Systems (Honor Track)**

**2024 Spring**

**Head TA, Operating Systems (Honor Track)**

**2023 Spring**

**TA, Introduction to Computing**

**2022 Fall**

**Head TA, Operating Systems (Honor Track)**

**2022 Spring**

**TA, Introduction to Computer System**

**2020 Fall**

## **SERVICES**

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Shadow Program Committee

Eurosys 2025

Artifacts Evaluation Committee

OSDI/ATC 2024

Artifacts Evaluation Committee

SIGCOMM 2024

Artifacts Evaluation Committee

OSDI/ATC 2023

## AWARDS & HONORS

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ByteDance Scholarship

2025

President's Scholarship of Peking University

2025

Merit Student of Peking University

2024

Outstanding Graduate of Peking University

2022

Third Prize Scholarship of Peking University

2020

Merit Student of Peking University

2019

Tian Chuang Scholarship

2019

Zhongying Tang Scholarship

2019

Merit Student of Peking University

2018